CN2022s final: Chap 1-Chap6, 7 problems in 4 pages, 100 points

Note:

1. This is an Online exam., open-book, open-everything. But, you should do it alone and cannot get helps from others.
2. Please upload your answer in one file (only .doc (MS word) or .pdf file is allowed). Be sure to have your name in the first page of your upload file.

1. (10%) The network default gateway applying to a host by DHCP is 192.168.5.33/25. Which option is the valid IP address of this host? Choose the right answer(s)(multiple options are possible). And give a short description about why?

(a). 192.168.5.155
(b). 192.168.5.47
(c). 192.168.6.40
(d). 192.168.5.132
(e). 192.168.5.114

2. (10%) efficiency of slotted ALOHA protocol

(a). Suppose N nodes with many frames to send, each transmits in slot with probability ***p***. Explain that the probability of a given node has success to send in a slot is $p(1-p)^{N-1}$. (2%)

(b). Continue from (a), show that the probability that any node has success to send in a slot is $f(p; N)= Np(1-p)^{N-1}$. (2%)

(c). Let p* is the value to maximizes $f(p; N)= Np(1-p)^{N-1}$. Show that $p^*=1/N$. (hint: find the p s.t. $\partial f(p; N)/\partial p = 0$) (2%)

(d). Continue from above (c), show that $\lim_{N\to\infty} f(p^*; N) = e^{-1}$ =0.3679, where $p^*=1/N$. (hint: $\lim_{n\to\infty}\left(1-\frac{1}{n}\right)^n = e^{-1}$) (2%)

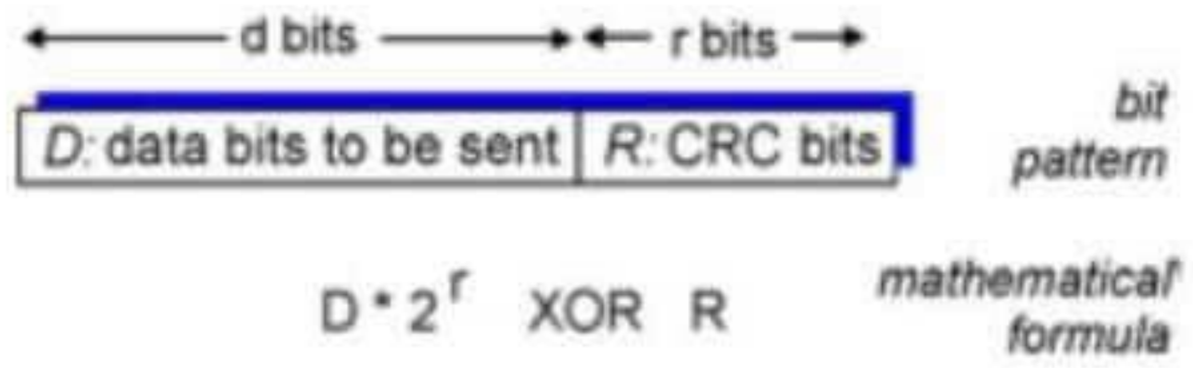
(e). Based on above, for un-slotted ALOHA (a.k.a pure ALOHA) protocol, explain that the probability of a given node has success to send in a slot is $p(1-p)^{N-1}(1-p)^{N-1}$. (2%)

3. (10%) Internet check-sum

Given a sequent of 12 bytes as (0x01, 0x02, 0x03, 0x04,0x05, 0x06,0x07,0x08, 0x09, 0x0A, 0xFF, 0xFF). What is the checksum number of this 12-byte network segment, according to the internet checksum algorithm?

4. **(10%)** CRC checksum:

   (a). Please explain how CRC checksum works in principle. (hint: R=remainder[$\frac{D \cdot 2^r}{G}$]) (4%)



   (b). Given, r=5, G=100111, D=10111011, what is R? (6%)

5. **(25%)** Suppose one day you bring your notebook to sit in the Computer Network course in our department, you power on your notebook and connected to our department wifi. Then you open a browse with domain name (https://tronclass.ntou.edu.tw) to login to the Tronclass course website, and download a course handout. Actually, from the moment you turn on your notebook up to you successfully download the course handout, many of network protocols have been involved between your notebook and different remote nodes. Assume all the network related information cached in your notebook (e.g. arp table, dns resolving table) are empty, and your notebook utilizes dhcp to acquire an IP address.

   (a). (16%) From the following network protocols,

   http, smtp, pop3, imap, dns, rtp, Skype, dhcp, arp, ospf/rip,

   please pick four of them which are very likely involved in the above scenario from the time power on up to the moment the handout is download. For each of the four protocols you choose, you should give the following information:1. The protocol name (what), 2. When it is involved (when), 3. Whom (the remote node) does the notebook communicate with (whom), and 4. Why it should be involved? (why). For example,

   1. name: wifi (802.11)
   2. when: the time it talks to remote AP
   3. whom: the remote AP
   4. why: it provides a wireless communication link with remote AP.

   (hint: you need both network address and mac address to communicate with other internet devices)

   (b) (3%) According to (a), when IP packet will be sent at first time? Explain your answer.

   (c) (3%) According to (a), when UDP packet will be sent at first time? Explain your answer.

(d) (3%) According to (a), when TCP packet will be sent at first time? Explain your answer.

6. **(30%)** Lab:Packet analysis

Figure 1(a) shows an Ethernet frame received, which carries an IP packet as the frame data payload. And, the IP packet embeds an TCP packet. The frame data marked with blue color (that is the byte stream "45 00 00 28 24 92 …c0 a8 00 64".) is the IP head part, and those followed is TCP packet. Figure 1(b) shows IPv4 packet format. For IPv4, both ver and head length fields take 4 bit, and the flgs field take 3 bits.

(a). Ethernet frame header part (6%)

1. Please check with Ethernet frame format (you may check from book or internet) to get what is the source MAC address carried in this frame packet? (2%)
2. Continue from above, what is the destination MAC address? (2%)
3. Continue from above, how did we know this data frame carries an IP packet in its frame data part? (hint: check the ***type*** field in the Ethernet header) (2%)

(b). IP packet header part (16%)

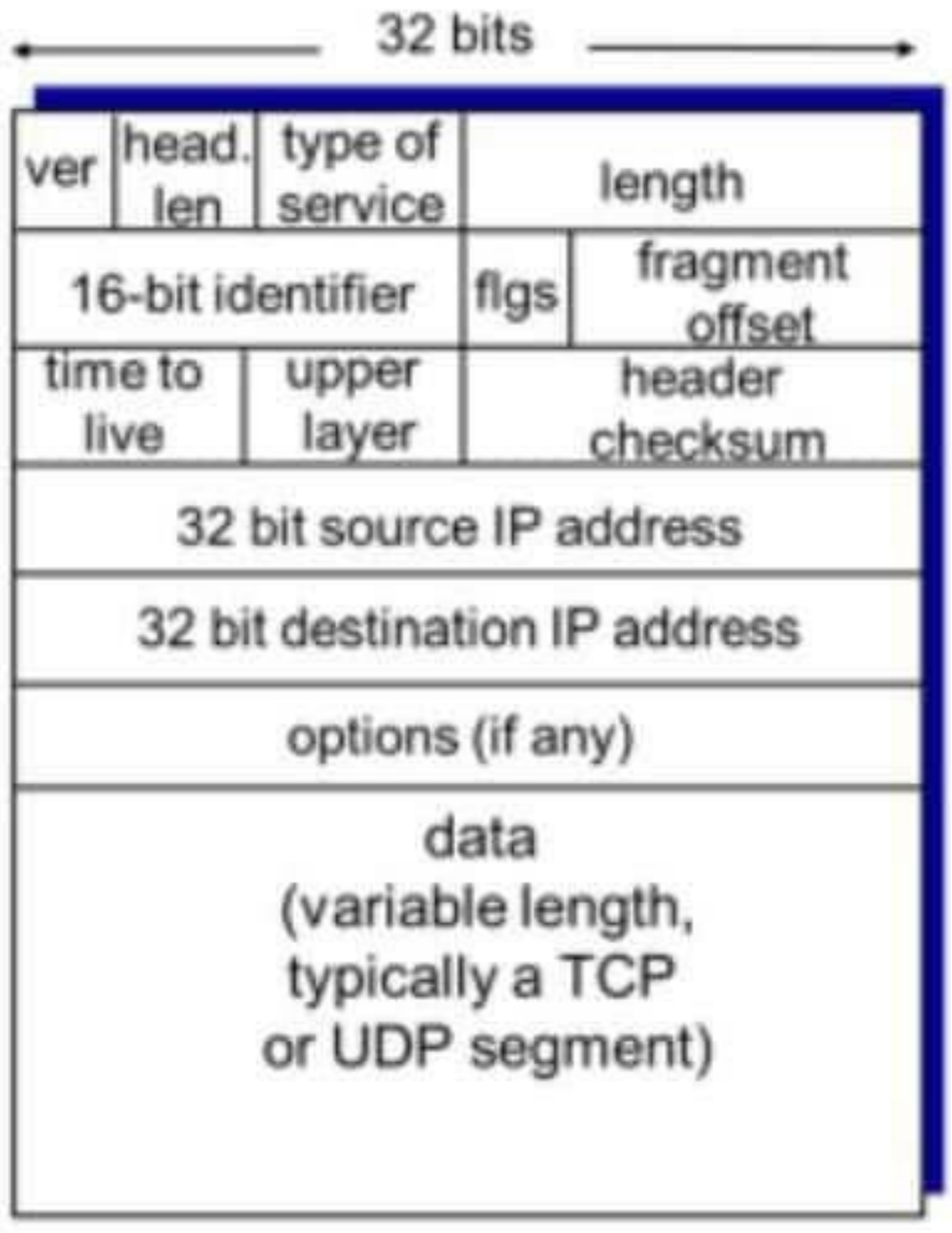
1. How should we know this packet is IPv4 but not IPv6 or others? (2%)
2. What is packet header length of this IPv4 packet? (2%)
3. What is total packet length of this IPv4 packet? (2%)
4. What is the TTL(time to live) value of this IPv4 packet? (2%)
5. What is the header checksum value of this IPv4 packet? (2%)
6. How did we know this IP packet carries an TCP datagram in its data part? (hint: check the ***upper-layer*** field in the IP header) (2%)
7. What is the source IP address carried in this IP packet? (2%)
8. Is this IPv4 packet contains IP header optional part? Why? (2%)

(c). TCP packet part (Please check with TCP packet format from book or internet) (8%):

1. What is the source port and destination port number carried in the TCP header? (hint: check the ***TCP*** packet format) (2%)
2. What is the TCP header Length of this TCP packet? (2%)
3. What TCP control bits (URG/ACK/PSH/RST/SYN/FIN) are set in this TCP packet? (2%)
4. How many TCP data bytes are carried in this TCP packet?

```
0000  f0 2f 74 00 15 7d 18 a6  f7 ab c7 0e 08 00 45 00   ·/t··}·· ······E·
0010  00 28 24 92 00 00 79 06  0c 2a ac d9 a3 2e c0 a8   ·($···y· ·*···.··
0020  00 64 01 bb f2 9f 1d 85  9b c6 65 c7 45 d7 50 11   ·d······ ··e·E·P·
0030  01 09 44 71 00 00 00 00  00 00 00 00               ··Dq···· ····
```

(a)



(b)

Figure 1. (a) Example of an IPv4 packet encapsulated in an Ethernet frame. (b)IPv4 packet format. (for Problem 5)

7. **(5%)** course feedback.
To improve course quality, please give your comments (especially for online course part) if any. Any comments are welcome. No matter what you write, you'll get full score of this question. If you don't know what to write, you can simply write done some short words, like "OK" or "no comment". Thanks!!

End of Questions.
Thanks for having you with us in this course. Happy Summer Break!!

5 層網路架構：

Top ↓ Down

- 應用層 Application Layer：負責支援各類網路應用程式，如 DNS、SMTP、HTTP 等。
- 傳輸層 Transport Layer：負責 Process 和 Process 間的資料傳輸，如 TCP、UDP 等。
- 網路層 Network Layer：負責將資料從來源轉送到目標位址
- 連接層 Link Layer：負責相鄰網路設備之間的資料傳輸
- 物理層 Physical Layer：負責在物理媒介上傳送位元，處理實際電信號。

1、DHCP Default Gateway：192.168.5.33/25

32−25＝7

$2^7=128$ → subnet mask：255.255.255.128

→ Default mask：/24

mean → $\overbrace{\underbrace{11111111}_{255}.\underbrace{11111111}_{255}.\underbrace{11111111}_{255}}^{24\text{ Bits}}.\overbrace{00000000}^{主机保留}$ ⇒ 那麼，192.168.5.0 ～ 192.168.5.255 这 256 个 IP 確實都在同網段中！

→ In This Case：/25

多跟主机保留区借了一位元來用。

mean → $\overbrace{\underbrace{11111111}_{255}.\underbrace{11111111}_{255}.\underbrace{11111111}_{255}.1}^{25\text{ Bits}}\overbrace{0000000}^{主机保留}$ ⇒

影響：192.168.5.X 被分成兩个網段

網段一 → 192.168.5.0 ～ 192.168.5.127 ← 192.168.5.33 is Here (DHCP Default Gateway)

網段二 → 192.168.5.128 ～ 192.168.5.255

Ans：(b)、(e)

因為 subnet mask 設定為 /25 的關係，只有 (b)、(e) 和 192.168.5.33 在同網段中，(a)、(e) 雖然也是掛著 192.168.5 開頭，但本質上已分屬不同網段。

2.

ALOHA Network：

Pure ALOHA：

- 隨時發送：有人要發訊息隨時傳
- 發生碰撞：若發生碰撞（沒收到 ACK），送端隨机等一段時間重送

→ 碰撞率極高，極限效率只有 18%

Slotted ALOHA：

時間被分割為大小相當的時槽（Slotted），每個 Slotted 的長度剛好為傳送一個 frame 所要的時間。

所有 client 共用同个基準時間，client 只能在 Slotted 的开始傳 frame。

→ 減少碰撞率，極限效率提高到 37%

(a) 解釋特定節点在一个 Slotted 傳送成功的机率是 $p(1-p)^{N-1}$

- 1 node want to send packet：p
- 其他所有 N−1 个節点都必須沒傳東西，單 node 不傳的机率是 1−p

→ $p\times(1-p)^{N-1}$

(b) Proof 任何 node 在一个 Slotted 傳送成功的机率是 $f(p,N)=Np(1-p)^{N-1}$

- 同 Slotted 不可能有 2 个以上 node 傳成功 ⟶ N 个 node 傳成功的机率是互斥事件
- Any Nodes 在一個 Slotted 內傳成功 ⟶ 剛好有个在 node 在一 Slotted 內傳成功！ ⟶ 把 N 種互斥 Case 相乘。

$f(p,N)=N\times p(1-p)^{N-1}$

(c) 設 p^* 是使 $f(p,N)=Np(1-p)^{N-1}$ 最大值。證 $p^*=\frac{1}{N}$ (find $\frac{\partial f(p,N)}{\partial p}=0$ 的 p)

→ 用一階导數＝0 來求解

$f(p)=Np(1-p)^{N-1}$，$\frac{\partial f}{\partial p}=N(1-p)^{N-1}+Np(N-1)(1-p)^{N-2}(-1)$

$N(1-p)^{N-2}\cdot[(1-p)-p(N-1)]=0$

$1-p-p(N-1)=0$，$1-p-Np+p=0$，$1=Np$ → $p^*=\frac{1}{N}$

(d) 證 $\lim_{N\to\infty}f(p^*,N)=e^{-1}=0.3679$，$p^*=\frac{1}{N}$ $\left(\lim_{n\to\infty}(1-\frac{1}{n})^n=e^{-1}\right)$

$f(\frac{1}{N},N)=N\times\frac{1}{N}\times(1-\frac{1}{N})^{N-1}=(1-\frac{1}{N})^{N-1}$

$\lim_{n\to\infty}(1-\frac{1}{N})^{N-1}=\lim_{n\to\infty}\frac{(1-\frac{1}{N})^N}{(1-\frac{1}{N})}=\frac{e^{-1}}{1}=e^{-1}\approx0.3679$

(e) 对於 Pure ALOHA，解釋特定節点在一个 Slotted 傳送成功的机率 $p(1-p)^{N-1}(1-p)^{N-1}$

→ In Pure ALOHA，沒有時間同步的問題，有封包就直接傳

→ Assume 1 Node 在 t_0 時傳封包，那麼：

不能跟前一封包重疊（其他 node 不能在 $[t_0-1,t_0]$ 区間発封包）：$(1-p)^{N-1}$

不能跟後一封包重疊（其他 node 不能在 $[t_0,t_0+1]$ 区間発封包）：$(1-p)^{N-1}$

特定一 node 發送机率：p

→ $p\times(1-p)^{N-1}\times(1-p)^{N-1}=p\times(1-p)^{2N-2}$

3、Internet Checksum

給定一 12 byte 序列：0x01 0x02 0x03 0x04 0x05 0x06 0x07 0x08 0x09 0x0A 0x0F 0x0F

→ 將序列兩兩一組合併：

0x0102, 0x0304, 0x0506, 0x0708, 0x090A, 0xFFFF } 加法合併

```
  0x0102        0x0506        0x1015        0x191D         0xFFFF
+ 0x0304      + 0x0708      + 0x090A      + 0x0001       - 0x191E
--------      --------      --------      --------       --------
  0x0406        0x0C0E        0x191E        0x191E         0xE6E1

  0x0708        0x0406        0x191E
+ 0x090A        0x0C0E      + 0xFFFF
--------      --------      --------
  0x1012        0x1015        0x1191D
```

↳ 超过 16 bit，加回

└→ 取補（減去 0xFFFF）

Ans：Check Sum = 0xE6E1

A 10
B 11
C 12
D 13
E 14
F 15

4、CRC Checksum

(a)
- 送端与收端先共同約定一个 r+1 位元的生成多項式 G
- 送端要送 d 位元的资料 D 時，会將 D 左移 r 尾元並補 0（$D \times 2^r$）
- 送端拿 $D \times 2^r$ 去除 G（每次操作用 XOR 而非減號），其餘數即為長 r 位元的 CRC Checksum R）
- 送端傳的時候会把 R、D 一起傳，收端接到資料後，一樣用 G 去除。
  - ↳ 若 Reminder = 0，結果正確
  - ↳ 若 Reminder ≠ 0，资料異常，重傳！

(b) r = 5，G = 100111，D = 1011011

对 D left shift 5 Bit：1011011 00000

做 XOR 除法：

```
              1 0 1 0 0 0 0 0
100111 / 1 0 1 1 0 1 1 0 0 0 0 0
       ⊕ 1 0 0 1 1 1
         -----------
             1 0 0 1 1 1
             1 0 0 1 1 1
             -----------
                   0 0 0 0 0 ← R
```

R = 00000

資料平面 Data Plane v.s. 控制平面 Control Plane

→ Network 重要構成

Data Plane：
- 主要負責「轉送」Forwarding
- 將抵達 Router 輸入端的資料報（Datagram）移动到適當的 Router 輸出端
- 單一 Router 内的局部功能
- 通常由硬体來負責執行，運作時間落在奈秒级（NanoSecond）

Control Plane：
- 主要負責「路由」Routing
- 規劃 & 決定 Packet 從來源到目標的完整路徑。
- 全域範圍（Network-wide）的邏輯決策
- 通常由軟体來負責執行，運作時間落在毫秒级（MircoSecond）

軟体控制網路 Software Defined Network SDN

傳統 將 Control Plane & Data Plane 整合在一同 Router 中，每台 Router 各自執行自身的 Routing Algorithm

SDN 將 Control Plane 從实体 Router 中抽離，改由遠端控制器（Remote SDN Controller）來統一計算，並將计算好的轉送表安裝到各 Router.

南向 API（Southbound API）：（SDN Controller 控制底層 Switch）SDN Controller 与其下方向的 Data Plane 通信の介面。Controller 利用南向 API 來管理底層设備，並將计算好的轉送規則安裝到 Switch 中。

北向 API（Northbound API）：（外部 App 控制 SDN Controller）SDN Controller 与其上方網路控制應用程式互动の介面。Network-Control Apps 是控制網路的中枢，其可透过北向 API 存取控制器所維護的網路全域狀態資讯。

SDN Core Component：

資料平面交換器 Data Plane Switches：硬体交换器，做 Data Plane 的 forwarding。根據 SDN Controller 發來的流程表處理 Packet；透过南向 API（Southbound API）和 SDN Controller 通訊。

SDN 控制器 SDN Controller：負責 maintenance 整個網路的全域狀態資訊。向下透过 Southbound API 管理底層 Switch，向上透过 Northbound API 提供服務和介面給上層 Application.

網路控制應用程式 Network-Control Apps：可利用 SDN Controller 提供的 API 實作各種網路功能。如 Routing、Load balance 等。

Why USE SDN？
- 讓 ISP 管理網路更方便
- Programmable Network 透过集中式 Controller 來計算和派発轉送表。
- 開放網路介面，允许第三方开発軟体，不再被製造商綁死

OSPF (Open Shortest Path First) → 负责單一自治系統(AS)，為了讓所有人都有完整的 Network topology，AS中的每个router都能將自己的連線狀態廣告「廣播」(flood) 給整个AS的其他 router。

*封裝在 ip datagram 中，而非TCP or UDP

→ Support 階層式 OSPF，為解決大型網路擴展性Problem

→ 这時 OSPF 会分成本地(local) 和 骨幹(backbone)

→ 連線狀態廣告只会在同区域內 broadcast.

BGP (Border Gateway Protocol) → 跨AS的協定，負責不同AS間的連線。

↳ 会建立半永久TCP Connection 來 exchange information.

GOAL：讓 Subnet 能向其他AS宣告自己的存在和可抵性。

eBGP (外部BGP)：跨AS間通訊，從相鄰AS獲取 Subnet 可抵性。

iBGP (內部BGP)：AS內的通訊，將eBGP獲取的可抵性資訊傳播給同一AS中的所有 router

Link Layer frame 封裝：

- 將原始 Datagram 包起來，加上 Header (標頭) 和 Trailer (結尾)
- frame 的 header 中：寫入來源 & 目標的 MAC Address
- 通常会一併封裝 Error Checking Bits 防止 Physical Layer 傳輸中發生錯誤或干擾 (e.g. CRC)

TCP 解多工 4-tuple 資訊： Source IP、Soure Port、Destination IP、Destation Porp